

PERSONAL INFORMATION

Davide Murari


✉ dm2011@cam.ac.uk

🌐 <https://www.davidemurari.com>

in <https://www.linkedin.com/in/davidemurari/>

PERSONAL STATEMENT

Postdoctoral Research Associate exploring the interplay between neural networks, dynamical systems, and structure-preserving numerical analysis.

PUBLICATIONS

The papers listed here and in the preprint section are those to which I have substantially contributed. They follow different author-ordering conventions depending on the venue. **The year on the left is written in magenta for those where I am the lead author. Joint lead authorship is indicated by a superscript [†].**

- 2026 Liu, Chaoyu, **Davide Murari**, Chris Budd, Lihao Liu, and Carola-Bibiane Schönlieb. "Enhancing Fourier Neural Operators with Local Spatial Features." SIAM Journal on Scientific Computing (2026). In press.
- 2026 **Murari, Davide**, and Nicola Sansonetto. "From Euler–Jacobi to Bogoyavlenskij and back." *Nonlinearity* 39, no. 1 (2026): 015016.
- 2025 **Murari, Davide**, Takashi Furuya, and Carola-Bibiane Schönlieb. "Approximation theory for 1-Lipschitz ResNets." In *The Thirty-ninth Annual Conference on Neural Information Processing Systems* (NeurIPS 2025), 2025.
- 2025 Celledoni, Elena, James Jackaman, **Davide Murari**, and Brynjulf Owren. "Predictions Based on Pixel Data: Insights from PDEs and Finite Differences." *Journal of Computational Physics* (2025): 114166.
- 2025 Celledoni, Elena, Ergys Çokaj, Andrea Leone, Sigrid Leyendecker, **Davide Murari**, Brynjulf Owren, Rodrigo T. Sato Martín de Almagro, and Martina Stavole. "Neural networks for the approximation of Euler's elastica." *Computer Methods in Applied Mechanics and Engineering* 435 (2025): 117584.
- 2024 Eliasof, Moshe[†], **Davide Murari**[†], Ferdia Sherry[†], and Carola-Bibiane Schönlieb. "Resilient Graph Neural Networks: A Coupled Dynamical Systems Approach." *ECAI - 27TH European Conference on Artificial Intelligence* (2024): 1607-1614.
- 2024 Sherry, Ferdia, Elena Celledoni, Matthias J. Ehrhardt, **Davide Murari**, Brynjulf Owren, and Carola-Bibiane Schönlieb. "Designing stable neural networks using convex analysis and ODEs." *Physica D: Nonlinear Phenomena* 463 (2024): 134159.
- 2023 Celledoni, Elena, **Davide Murari**, Brynjulf Owren, Carola-Bibiane Schönlieb, and Ferdia Sherry. "Dynamical Systems–Based Neural Networks." *SIAM Journal on Scientific Computing* 45, no. 6 (2023): A3071-A3094.
- 2023 Celledoni, Elena, Andrea Leone, **Davide Murari**, and Brynjulf Owren. "Learning Hamiltonians of constrained mechanical systems." *Journal of Computational and Applied Mathematics* 417 (2023): 114608.
- 2022 Celledoni, Elena, Ergys Çokaj, Andrea Leone, **Davide Murari**, and Brynjulf Owren. "Lie group integrators for mechanical systems." *International Journal of Computer Mathematics* 99, no. 1 (2022): 58-88.

MAIN PREPRINTS

- 2026 Furuya, Takashi[†], **Davide Murari**[†], Carola-Bibiane Schönlieb. "Approximation Theory for Lipschitz Continuous Transformers." arXiv preprint arXiv:2602.15503 (2026).
- 2026 Massucco, Alex[†], **Davide Murari**[†], Carola-Bibiane Schönlieb. "Deep Network Trainability via Persistent Subspace Orthogonality." arXiv preprint arXiv:2508.02882 (2026).
- 2025 De Marinis, Arturo, **Davide Murari**, Elena Celledoni, Nicola Guglielmi, Brynjulf Owren, and Francesco Tudisco. "Approximation properties of neural ODEs." arXiv preprint arXiv:2503.15696 (2025).

- 2024** Canizares, Priscilla, **Davide Murari**, Carola-Bibiane Schönlieb, Ferdia Sherry, and Zakhar Shumaylov. "[Symplectic Neural Flows for Modeling and Discovery](#)." arXiv preprint arXiv:2412.16787 (2024).
- 2024** Betcke, Marta M., Lisa Maria Kreusser, and **Davide Murari**. "[Parallel-in-Time Solutions with Random Projection Neural Networks](#)." arXiv preprint arXiv:2408.09756 (2024).

RESEARCH POSITIONS

1st October 2024 - 31st July 2028

Postdoctoral Research Associate in the Cambridge Image Analysis group of Professor Carola-Bibiane Schönlieb, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. The position is funded by two EPSRC programme grants under the projects EP/V026259/1 and EP/Y028783/1.

RESEARCH SECONDMENTS

March-May 2023

Visiting researcher at the Isaac Newton Institute for Mathematical Sciences, Cambridge, United Kingdom, during the thematic programme titled "The mathematical and statistical foundation of future data-driven engineering".

October 2021

Visiting researcher at the Isaac Newton Institute for Mathematical Sciences, Cambridge, United Kingdom, during the thematic programme titled "Mathematics of deep learning".

EDUCATION

2020-September 2024

PhD in Numerical Analysis, Norwegian University of Science and Technology, supervised by Professors Elena Celledoni and Brynjulf Owren. Thesis title: "[Neural Networks, Differential Equations, and Structure Preservation](#)".

2018-2020

Master in Mathematics (L-40), University of Verona, Italy. Thesis title: "Integrable Non-Hamiltonian Systems: From B-Integrability to Euler-Jacobi Theorem and Back", supervised by Professor Nicola Sansonetto. Final mark: 110/110 cum laude.

2015-2018

Bachelor in Applied Mathematics (L-35), University of Verona, Italy. Thesis title translating to "An Overview of Ergodic Theory and Dynamical Billiards", supervised by Professor Nicola Sansonetto. Final mark: 110/110 cum laude.

PEER REVIEW

Reviewer for the following scientific journals:

- [Calcolo](#)
- [Journal of Computational and Applied Mathematics](#)
- [IMA Journal of Applied Mathematics](#)
- [IMA Journal of Numerical Analysis](#)
- [SIAM Journal of Scientific Computing](#)
- [SIAM Journal on Mathematics of Data Science](#)
- [Mathematics of Computation](#)
- [Journal of Computational Physics](#)

ORGANISED EVENTS

Co-organiser of a summer school and a conference (upcoming):

- [SCML 2026: International Conference on Scientific Computing and Machine Learning](#), 14–17 September 2026, University of Bath.
- Summer school on the "Mathematical Foundations of Deep Learning", 3–7 August 2026, Centre for Mathematical Sciences, University of Cambridge.

ORGANISED MINI-SYMPOSIA

Co-organiser of three mini-symposia:

- Mini-symposium at SciCADE 2026 (upcoming), *Improving numerical methods with machine learning*, Edinburgh, 26 June–3 July 2026.
- Mini-symposium at SciCADE 2024, *Dynamical systems, structure preservation and deep learning*, Singapore, 15–19 July 2024.
- Mini-symposium at Geometric Science of Information (GSI), *Deep Learning: Methods, Analysis and Applications to Mechanical systems*, St. Malo, France, 30 August–1 September 2023.

EDITORIAL WORK

2021-2022 Guest editor for the ECMI Annual reports of the years 2021 and 2022.

SELECTED FUNDING SOURCES

January 2024 Trond Mohn Foundation (TMS) grants: Support for attending the Geilo Winter School "Graphs and Applications", from the 21st to the 26th of January 2024. The funding has been of 12,000.00 NOK.

March-April 2023 Simons visiting fellowship: Support for participating in the thematic semester "The mathematical and statistical foundation of future data-driven engineering" during the days from the 7th of March 2023 to the 2nd of May 2023. The funding has been £1,400.00.

January 2023 Trond Mohn Foundation (TMS) grants: Support for hosting Professor Nicola Sansonetto, a collaborator, at our department during the week from the 15th to the 21st of January 2023. The funding has been of 10,000.00 NOK.

October 2021 Isaac Newton Institute: Support for participating in the thematic semester titled "Mathematics of deep learning" during the days from the 8th to the 30th of October 2021. The funding has been £330.00.

TEACHING AND SUPERVISION

10 November - 14 November 2025 Teaching the master-level mini-course "[Neural Networks as Dynamical Systems](#)" at the University of Verona.

29 September - 03 October 2025 Invited lecturer at the [11th CISM-AIMETA Advanced Course on "Machine Learning for Solid Mechanics"](#).

July 2025 - September 2025 Supervision of two summer interns at the University of Cambridge through the Philippa Fawcett Internship and Summer Research in Maths (SRIM) programmes.

February 2025 - December 2025 Co-supervision of the master's thesis of a student from Politecnico di Milano.

January - March 2025 Teaching the graduate course "[Geometric Numerical Analysis and Deep Learning](#)" at the University of Cambridge.

November 2024 Teaching the master-level mini-course "[Geometric Methods for Differential Equations and Learning Applications](#)" at the University of Verona.

Fall 2023 Assistant for the course "Numerical Solution of Partial Differential Equations Using Element Methods - TMA4220" at NTNU.

Fall 2022 Assistant for the course "Numerical Solution of Partial Differential Equations Using Element Methods - TMA4220" at NTNU.

Spring 2022 Assistant for the course "Optimization 1 - TMA4180" at NTNU.

Fall 2021 Assistant for the course "Numerical Solution of Partial Differential Equations Using Element Methods - TMA4220" at NTNU.

Spring 2021 Assistant for the course "Differential Equations and Dynamical Systems - TMA4165" at NTNU.

SELECTED INVITED TALKS

July 2025 Joint SIAM/CAIMS annual meeting (AN25), Montreal, Canada. Talk title: "Symplectic Neural Flows".

October 2024 Workshop "Deep Learning for PDE-based Inverse Problems", Oberwolfach, Germany. Talk title: "Dynamical systems-based structured networks".

October 2024 SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, USA. Talk title: "Structure-Preserving Solutions of Hamiltonian Systems Based on Neural Networks"

February 2024 BIRS Workshop "Structured Machine Learning and Time-Stepping for Dynamical Systems", Banff, Canada. Talk title: "Improving the robustness of Graph Neural Networks with coupled dynamical systems".

September 2023 TES Conference on Mathematical Optimization for Machine Learning, Berlin, Germany. Talk title: "Predictions Based on Pixel Data".

August 2023 ICIAM 2023, Tokyo, Japan. Talk title: "Structured neural networks and some applications".

June 2023 ECMI Conference 2023, Wroclaw, Poland. Talk title: "Learning Hamiltonians of constrained mechanical systems".

June 2023 FoCM 2023, Paris, France. Talk title: "Structured neural networks and some applications to dynamical systems".

February 2023 SIAM CSE, Amsterdam, Netherlands. Talk title: "Structured neural networks and their relevance for mechanical systems".

December 2022 Theoretical and Computational aspects of Dynamical Systems (HB60), Trysil, Norway. Talk title: "Dynamical systems-based neural networks".

July 2022 SciCADE, Reykjavík, Iceland. Talk title: "Structure preserving neural networks coming from ODE models".

May 2022 Machine Learning and Dynamical Systems Seminar of the Alan Turing Institute, Online. Talk title: "Learning Hamiltonians of constrained mechanical systems".

HOBBIES

Interested in communicating mathematics, sharing content about various topics in this field with an ongoing outreach project:

- in written format on the newsletter: [Substack newsletter](#)
- in video format on the YouTube channel: [Mathone | Davide Murari](#) (8.7k subscribers).

Long-distance runner; completed multiple half-marathons and four marathons.

LANGUAGE SKILLS

Mother tongue Italian

Other languages

English Full professional proficiency

Norwegian (Bokmål) Intermediate